Single Microcomb Source for Ultra-Scalable Datacenters for Dense Deep Neural Network Workloads

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Abstract: We propose an optical datacenter network architecture for artificial intelligence, leveraging a single microcomb, with comb-lines distributed depending on the workload. The microcomb can support 120 Pbit/s and in excess of one million GPUs. ©2025 The Author(s)

1. Introduction

Machine learning applications introduce increasingly stringent performance and efficiency requirements in datacenters (DCs). Optical DC networks, based on optical circuit switches have recently received attention, not least when utilising optical frequency combs (OFCs) [1-5]. In [1] a comb is used instead of a laser bank in a wavelength-band switching experiment. In [2], the reduced power benefits of a comb source over a laser bank is discussed in an optical switching scenario aiming for 100,000 nodes in a DC, and in [3] fast silicon switches are demonstrated using fixed wavelength lasers. [4] uses a comb and sub-ns switches, and discusses sharing a single comb source between 64 racks, where [5] proposes a comb synchronization layer, and [6] proposes to use an external multi-color laser for integrated silicon modulators. In this paper, we propose a novel DC network architecture, relying on a single microcomb source, distributing light "as a commodity'' throughout the DC. We numerically investigate how the inter-server-bandwidths scale with the number of supported GPUs in this architecture, and we experimentally investigate the data capacity scaling of a single microcomb source. Specifically, we split an optical frequency comb in 4096 SDM paths, and demonstrate that the worst branch can support 29.45 Tbit/s in the C-band, corresponding to about 120 Pbit/s in total, supporting more than 1,000,000 GPUs with 1.7 Tbit/s inter-server bandwidth.

2. Network architecture

Our proposed architecture is illustrated in Fig. 1. The optical frequency comb from a single microcomb source is spatially distributed to a network of comb-distribution units (CDUs), each of which serves multiple racks. A CDU manages a full band (e.g. C and/or L-band) of WDM lines and administers lines to top-of-the-rack (ToR) switches upon request. When these WDM lines have been assigned, the ToRs can modulate them to transmit information. The modulated light is distributed across a network of Optical Circuit Switches (OCSs).

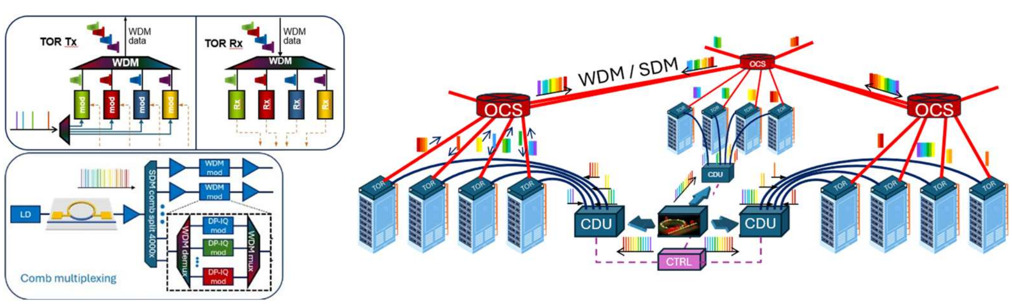

Fig. 1. Single comb-source based data centre network. Mid-left: A single comb-source has its output split in 4096 spatial paths to provide WDM channels to all racks in the DC. Right: The proposed network transmits identical WDM-combs to 4096 comb distribution units (CDUs), where requested WDM-lines are supplied to individual racks, data modulated, and sent into the network through optical circuit switches (OCS). The OCSs may be connected in a ring configuration and use WDM and SDM dimensions. Top left: The top-of the rack (TOR) switches will request a number of WDM-lines from the CDU corresponding to the data needed to be offloaded from the rack. These lines will be modulated with the rack-data, and sent onwards to the OCS. The TOR will also receive incoming data and distribute it to GPUs in the rack.

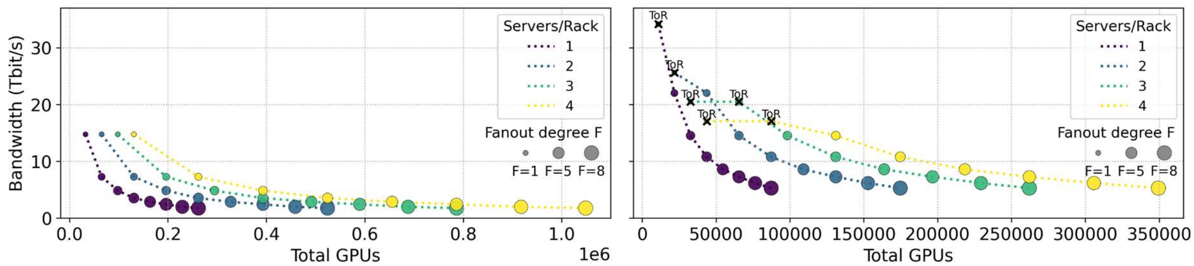

Fig. 2. Bandwidth between multi-GPU servers provided by a single optical frequency comb. We consider different non-overlapped (left) and overlapped (right) fan-out configurations, describing how CDUs are mapped to ToRs.

Large-scale deep neural network (DNN) training is among the most demanding workloads, requiring thousands of interconnected GPUs. We focus on dense DNN training, distributed according to the PTD-P best practice [7] and using a common hardware configuration of up to four multi-GPU servers in a rack, each with eight GPUs. PTD-P results in Tensor Parallel (TP) traffic remaining local within a server, while Data Parallel (DP) and Pipeline Parallel (PP) traffic can occur between servers. To match the inter-server communication pattern, we configure the OCS paths into a ToR-to-ToR ring. This ring configuration aligns with the AllReduce ring collective used for DP gradient synchronization. We primarily consider that PP is between servers of the same rack but note that PP traffic between servers (hosting sequential pipeline stages) can also be embedded on the ToR-to-ToR ring. These rings can be subdivided into smaller rings and reconfigured between training jobs, allowing efficient multi-tenant training. We model ToR switching capacity as 102.4Tbit/s, based on state-of-the-art switching ASICs and divided equally between ToR-to-ToR and intra-rack connections.

Fig. 2 shows the number of GPUs supported by a single optical frequency comb and resulting inter-server bandwidth. Each rack hosts up to four multi-GPU servers, noted by color. Marker size indicates the fan-out degree F or the number of ToRs per CDU; e.g. in Fig. 1 fan-out is F=4. In the non-overlapping fan-out configuration of Fig. 2 (left) each ToR is connected to at most one CDU. In this way the optical frequency comb is spread across thousands of racks, reaching over a million GPUs with 1.76Tbit/s between servers. To further increase the bandwidth, we consider a topology with overlapping CDU-to-ToR mapping, using SDM for optical paths. In Fig. 2 (right) each ToR is connected to 3 CDUs. This significantly decreases the spread in terms of supported GPUs, concentrating the available bandwidth. At higher bandwidths, ToR switching capacity becomes the limiting factor (shown with "x“ markers), particularly as intra-rack connectivity demands increase with more servers per rack. Even when adopting an optimistic estimate of 25,000 GPUs per datacenter building, this setup still spans multiple buildings. Our experimental results demonstrate that the WDM lines can traverse such multi-building distances (80 km, see below). Our topology can be extended to support further parallelism techniques that may require lower topology diameter or higher communication degree, utilizing the independently steerable WDM lines or reconfiguration of the OCS’s [8].

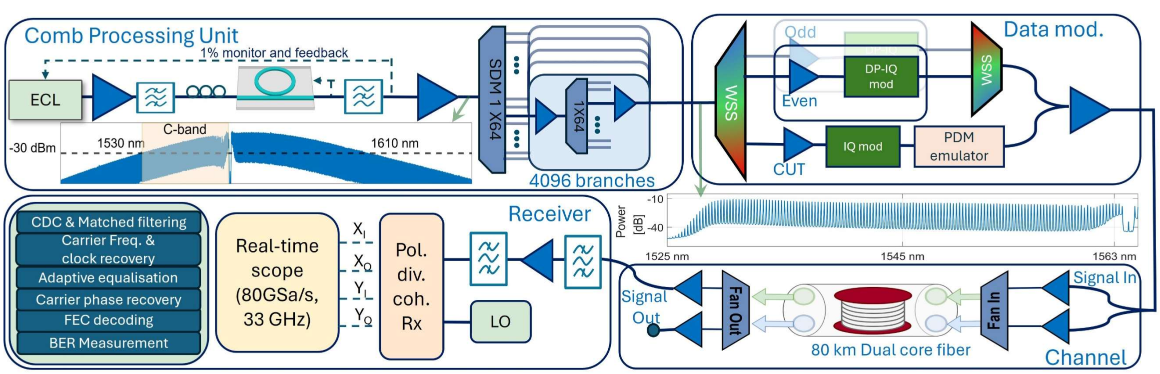

Fig. 3. Experimental setup. Comb Processing Unit: A tunable laser pumps a micro-ring resonator to produce comb lines in a bright soliton Kerr comb spectrum (shown in the inset). The comb is split to 4096 spatial channels. The comb spectrum before data modulation is shown in inset. Data modulation: The highest loss spatial channel is selected, and a channel under test (CUT) is chosen in a wavelength selective switch (WSS), and other wavelength channels are split in odd and even channels. The CUT is placed among the dummy channels using a second WSS, and scanned across the C-band. The signal is transmitted over 3 km SSMF and 80 km dual-core fiber to mimic intra-/inter-DC communication, and is subsequently received in a polarization diverse coherent receiver.

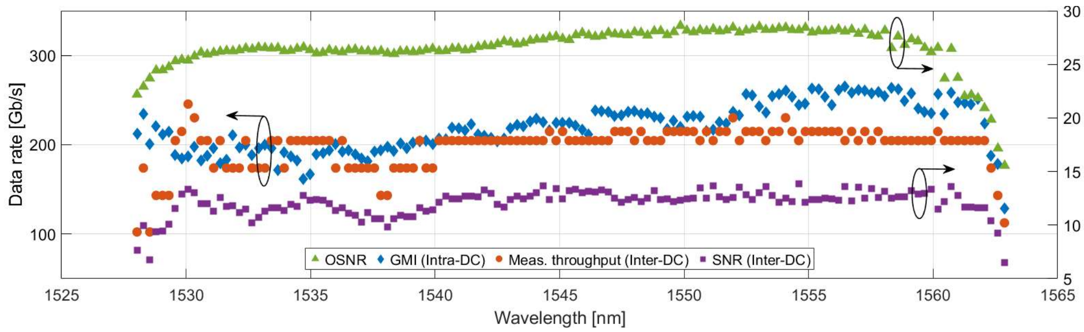

Fig. 4. Experimental results. Estimated OSNR after 4096 splits. Measured GMI for Intra-DC scenario, error-free data throughput and SNR in inter-DC scenario.

3. Experimental characterization

The experimental setup is shown in Fig. 3. An in-house fabricated SiN micro-resonator [9] creates the microcomb used in this work. The comb has a free spectral range (FSR) of 32.88 GHz, and an intrinsic quality factor of 15E6, and is a single soliton comb locked for long term stabilization [10]. The 28-dBm optical pump is at 1563 nm results in an output comb power of roughly 3.2 mW. Fig. 3 (inset) shows the microcomb output spectrum notch-filtering the pump. The comb is split into 4096 spatial copies by cascading two 1x64-splitters with a loss variation of less than 2-dB. The SDM channel with the highest loss is chosen. The C-band spectrum of the comb after SDM-splitting is shown in Fig. 3 (inset). Even after 4096 split and subsequent amplification, an average OSNR of 30 dB is estimated for most wavelength channels (shown in Fig 4, right y axis). To investigate an intra-DC scenario, the comb is modulated and transmitted 3 km, and for an inter-DC scenario, an 80 km dual-core fiber (Lightera MCF-ULL SCUBA 2X) is used. The co-propagation scheme, where the signals travel through the cores travel in the same direction is used. A total optical bandwidth of 4.37 THz, with a narrow guard band of 880 MHz. gave 134 WDM channels for data transmission. The data throughput of each WDM comb line is sequentially characterized as the CUT amongst all remaining channels, divided into even and odd individually modulated *dummy* channels.

The CUT is modulated up to 64-QAM, probabilistically shaped using a Maxwell-Boltzmann distribution at 32 Gbaud root-raised cosine pulse shape followed by a PDM emulator to mimic Dual-pol transmission. An inner 20% overhead LDPC code is used while an outer 1% overhead BCH code is assumed to correct any remaining errors after LDPC decoding (BER < 1E-5) [11]. Finally, a pilot overhead of 3% is subtracted to arrive at the throughput result. The measured OSNR for each WDM channels are shown in Fig. 4. The shown channels are error-free (blue, left axis), and from the measured SNR (right Y-axis) the estimated total GMI is 29.45 Tbit/s for intra-DC scenario and 31 Tbit/s for inter-DC scenario, giving a 220 Gbit/s and 231 Gbit/s for each wavelength channels for intra and inter-DC scenario respectively. The intra-DC scenario is currently impaired by equipment limitations. These results show the capability of a microcomb as the source for both intra and inter DC scenarios.

4. Conclusion

We have proposed a novel data centre network architecture taking advantage of a single optical frequency comb. Our model shows that this architecture can support multi-tenant DNN training with 1.7 Tbit/s inter-server bandwidth for over one million GPUs, and we have experimentally demonstrated the C-band performance of one branch of the 4096x spatial channels to support 29.45 Tbit/s, which corresponds to a total data capacity of 120 Pbit/s. The 4096-split branches may be equally used for intra-DC (3 km) and inter-DC (80 km) reac.

5. References

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